

Community Character Weighting: A Framework for Operationalizing Communities of Interest in Algorithmic Redistricting

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Abstract

The criterion “preserve communities of interest” appears in redistricting law across at least thirty states, yet courts that invoke it offer no quantitative definition. This paper establishes a mathematical framework that operationalizes communities of interest as cosine similarity of tract-level character vectors, yielding edge weight modifiers for the METIS graph partitioning engine at the core of the BISECT redistricting system. The framework is modular: any measurable tract-level signal—economic composition, land use, housing stock, commuting patterns, topographic setting, administrative zone co-membership, or transit accessibility—maps to a dimension of the character vector. The resulting edge weight formula $w(u, v) = w_{\text{base}} \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v)$ satisfies symmetry, non-negativity, and graceful degradation to the unweighted baseline when all tracts are identical. We ground the framework in the legal doctrine of communities of interest as articulated in *Reynolds v. Sims* (1964), *Shaw v. Reno* (1993), and the independent redistricting commission criteria enacted in California, Colorado, Arizona, and Washington. The framework is partisan-neutral: no electoral data, racial composition, or incumbency information enters any component of the weight computation. Papers M.1–M.7 implement specific instantiations of the framework; M.8 provides a composite index. This paper serves as the foundational theoretical reference for the entire M-track.

Keywords

Communities of interest, Redistricting, Edge weights, METIS, Cosine similarity, Community character, Algorithmic redistricting, Shaw v. Reno, Property-tax fiscal bond

1 Introduction

Every major redistricting reform of the past two decades has placed “communities of interest” on the list of criteria that map-drawers must satisfy or at least consider [National Conference of State Legislatures, 2021]. California’s Proposition 11 (2008), which created the Citizens Redistricting Commission, requires the Commission to “favor keeping together communities of interest”—a phrase that appears four times in the enabling statute. Colorado’s independent commission must give “communities of interest, including but not limited to ethnic, cultural, economic, trade area, geographic, and demographic factors” explicit weight under Art. V, §44 of the state constitution. Arizona’s Independent Redistricting Commission lists preservation of communities of interest among its six mandatory criteria, alongside equal population, compactness, and VRA compliance. Washington’s Redistricting Commission is directed by RCW 44.05.090 to give consideration to “communities of interest including racial and language minority groups.”

Yet courts that review maps drawn under these standards face an immediate problem: none of the statutes define what a community of interest is, how to measure whether one has been preserved, or how to adjudicate competing claims when communities of interest conflict with compactness or with county integrity. Judges must rely on the testimony of expert witnesses, who in turn rely on qualitative descriptions, anecdotal neighborhood histories, and—when challenged—on informal assertions that two neighborhoods share a common identity. No mathematical definition exists in case law. The result is that “preserve communities of interest” is legally recognized but operationally unenforceable in quantitative terms.

This paper proposes to close that gap. We define a community character vector $\mathbf{c}(t)$ for each census tract t , whose dimensions encode measurable signals of tract identity: the occupational composition of resident workers, the land-use mix, the housing tenure and vintage, the commuting orientation, the topographic setting, the set of administrative zones the tract belongs to, and the transit accessibility of its residents. Two adjacent tracts u and v that share similar character vectors are, by this definition, members of the same community; the cosine similarity of their character vectors provides a continuous measure of community

cohesion. We then define an edge weight modifier

$$w(u, v) = w_{\text{base}} \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v), \quad (1)$$

which, when passed to the METIS graph partitioner underlying BISECT, causes the partitioner to be geometrically reluctant to cut through edges connecting similar-character tracts. A cut that would split a community of interest becomes a costly cut; a cut that separates tracts of genuinely different character is cheap. The framework is partisan-neutral: no electoral data, no racial composition, and no incumbency information enters any component of the character vector or the weight formula.

1.1 Contribution

This paper makes the following contributions:

1. **Mathematical framework:** We provide the first precise mathematical definition of “community of interest” as cosine similarity of tract-level character vectors, formalized as an edge weight modifier for METIS graph partitioning (§2).
2. **Legal grounding:** We trace the communities of interest doctrine from *Reynolds v. Sims* (1964) through *Shaw v. Reno* (1993) and into modern state-level commission criteria, demonstrating that the framework satisfies the neutrality requirements imposed by Shaw (§3).
3. **Signal taxonomy:** We provide a structured catalogue of seven community character signals (M.1–M.7), each with its data source, similarity metric, availability, and legal precedent (§4).
4. **Integration specification:** We specify how M-track weights compose with existing BISECT edge weight modes via alpha-blending (§5).

1.2 Scope and Limitations

This paper is a framework paper. It does not report redistricting runs or district-level outcomes; those are the province of M.1–M.7, which validate each signal’s empirical effect on the plans produced by BISECT. This paper establishes the mathematical and legal scaffolding that all M-track papers share. The framework is also not a claim about partisan effect: because the character signals are partisan-neutral by construction, the framework does not predict or guarantee any particular partisan distribution of districts. Whether communities

of interest so defined produce more or less competitive districts, more or less proportional seat allocations, or better or worse compactness scores are empirical questions addressed in M.1–M.7 and M.8.

1.3 Paper Organization

Section 2 defines the mathematical framework. Section 3 grounds it in redistricting doctrine. Section 4 previews the seven signal types. Section 5 explains how M-track weights integrate with BISECT’s three-layer compositor. Section 6 concludes with a discussion of the framework’s role in expert witness testimony.

2 The General Framework

2.1 Character Vectors

Let $\mathcal{T}$ denote the set of census tracts in the state under redistricting. For each tract $t \in \mathcal{T}$, we define a *community character vector*

$$\mathbf{c}(t) = (c_1(t), c_2(t), \dots, c_k(t)) \in \mathbb{R}_{\geq 0}^k, \quad (2)$$

where each coordinate $c_i(t)$ is a non-negative real-valued signal derived from publicly available, partisan-neutral data. The number of dimensions k depends on which signals are active for the given plan configuration; any subset of M.1–M.7 may contribute one or more dimensions. We require only that each $c_i(t) \geq 0$ and that the vector is defined for every tract with nonzero population.

2.2 Cosine Similarity

Given two adjacent tracts $u, v \in \mathcal{T}$, their community similarity is

$$\text{sim}(\mathbf{c}_u, \mathbf{c}_v) = \frac{\mathbf{c}_u \cdot \mathbf{c}_v}{\|\mathbf{c}_u\| \|\mathbf{c}_v\|} \in [0, 1], \quad (3)$$

where the dot product and norms are the standard Euclidean ones. Because all character vector coordinates are non-negative, the cosine similarity is always in $[0, 1]$; negative similarity—which would arise from negative character signals and would have no natural interpretation—is impossible by construction.

Cosine similarity measures the *orientation* of the character vectors rather than their magnitude. This choice is deliberate: two tracts that are economically active in the same

mix of industries should be recognized as similar whether one is a large urban tract and the other is a small suburban one. Scaling by population or area is not done at the similarity level; if population weighting is desired for a particular signal, it enters $c_i(t)$ before the similarity computation, not after it.

2.3 Edge Weight Modifier

Let $G = (\mathcal{T}, E)$ be the tract adjacency graph, where an edge $(u, v) \in E$ exists when tracts u and v share a boundary of positive length (zero-length point adjacencies are excluded). The BISECT platform assigns each edge a base weight $w_{\text{base}}(u, v)$ determined by the active base weight mode (geographic boundary length, county co-membership, or unweighted). The community character modifier replaces or augments that base weight as

$$w(u, v) = w_{\text{base}}(u, v) \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v). \quad (4)$$

When $\text{sim}(\mathbf{c}_u, \mathbf{c}_v) = 1$ (identical character vectors), the weight is unchanged from w_{base} . When $\text{sim}(\mathbf{c}_u, \mathbf{c}_v) = 0$ (orthogonal character vectors, meaning completely different community profiles), the edge weight collapses to zero, making the cut between those tracts free. Intermediate similarities yield intermediate weights.

2.4 Mathematical Properties

Symmetry. $\text{sim}(\mathbf{c}_u, \mathbf{c}_v) = \text{sim}(\mathbf{c}_v, \mathbf{c}_u)$ for all pairs (u, v) , since the dot product is symmetric. Consequently $w(u, v) = w(v, u)$, which METIS requires: undirected graphs have symmetric edge weights.

Non-negativity. $\text{sim}(\mathbf{c}_u, \mathbf{c}_v) \geq 0$ whenever $\mathbf{c}_u, \mathbf{c}_v \in \mathbb{R}_{\geq 0}^k$, since the dot product of two non-negative vectors is non-negative. METIS requires non-negative integer edge weights; the implementation rounds $w(u, v)$ to the nearest positive integer after scaling, with a minimum value of 1.

Graceful degradation. If all tracts have identical character vectors—formally, if $\mathbf{c}(t) = \mathbf{c}$ for all t —then $\text{sim}(\mathbf{c}_u, \mathbf{c}_v) = 1$ for all edges, and $w(u, v) = w_{\text{base}}(u, v)$ for all edges. The community character modifier does not change the plan at all when there is no character variation to exploit. This ensures that the framework introduces no distortion in states where the chosen signal has no geographic variation.

Degenerate case: zero vector. If a tract t has $\mathbf{c}(t) = \mathbf{0}$ —for example, because no workers reside in a tract that is entirely commercial—the cosine similarity is undefined. The implementation handles this by substituting $\text{sim} = 1$ (treat as identical to any neighbor), which is conservative: it does not create a free cut but also does not create an artificially strong bond.

2.5 Well-Definedness of the Similarity

Cosine similarity is undefined when either vector is the zero vector. A tract with zero character in all dimensions is a tract that the available data cannot characterize; it should not bias the partitioner in either direction. The substitution $\text{sim} = 1$ for zero-vector tracts is equivalent to treating such tracts as perfectly similar to their neighbors in the character dimension, leaving their cut costs governed entirely by the base weight mode.

Additionally, for the administrative zone co-membership signal (M.6), similarity is not cosine-based but ratio-based (fraction of zones shared); see M.6 for the precise formula. The zone co-membership formula satisfies symmetry, $[0, 1]$ range, and graceful degradation independently, and is a valid plug-in replacement for $\text{sim}(\mathbf{c}_u, \mathbf{c}_v)$ in equation (4).

2.6 Invariants for Implementation

Any correct implementation of the framework must satisfy:

- **similarity_symmetric:** $\text{sim}(u, v) = \text{sim}(v, u)$ for all adjacent tract pairs.
- **similarity_self_is_one:** $\text{sim}(u, u) = 1.0$ for all tracts u (a tract is perfectly similar to itself).
- **weight_stays_nonneg:** $w(u, v) \geq 0$ for all edges after blending and rounding.
- **weight_bounded_above:** $w(u, v) \leq w_{\text{base}}(u, v)$ under the multiplicative formula (since $\text{sim} \leq 1$, so $w_{\text{base}} \times \text{sim} \leq w_{\text{base}}$). Note: M.6’s additive boost variant ($w = w_{\text{base}} \times (1 + \alpha \times \text{score})$) can exceed w_{base} and is a distinct variant of the framework; see M.6 §3 for reconciliation.

These are L0 unit test invariants checked on every build of the BISECT crate implementing this framework.

3 Legal Grounding

3.1 Origins of the Communities of Interest Doctrine

The constitutional foundation for population equality in legislative districts was established in *Reynolds v. Sims*, 377 U.S. 533 (1964) [U.S. Supreme Court, 1964]. Chief Justice Warren’s majority opinion held that the Equal Protection Clause requires that seats in both houses of a state legislature be apportioned on a population basis. But the Court also acknowledged that rigid adherence to mathematical equality would conflict with legitimate state policies of respecting existing political subdivisions:

“A State may legitimately desire to maintain the integrity of various political subdivisions, insofar as possible, and provide for compact districts of contiguous territory in apportioning its legislature.”

This passage planted the seed of the communities of interest doctrine. Political subdivisions—counties, cities, townships—are recognized as legitimate organizing units not only because they are administratively convenient but because they represent communities with shared interests in local governance. The doctrine later generalized beyond formal subdivisions to any group of residents who share economic, social, cultural, or civic interests.

3.2 *Shaw v. Reno* and the Neutrality Requirement

Shaw v. Reno, 509 U.S. 630 (1993) [U.S. Supreme Court, 1993], created the most important constraint on community-based redistricting: race cannot be the predominant factor in drawing district lines. Justice O’Connor’s majority opinion held that a district shaped primarily to aggregate racial groups is constitutionally suspect and triggers strict scrutiny under the Equal Protection Clause, even if the resulting district complies with the Voting Rights Act. The Court in *Miller v. Johnson*, 515 U.S. 900 (1995) [U.S. Supreme Court, 1995], clarified that the racial predominance test applies whenever race is the “overriding, predominant force” in the line-drawing process, not merely when districts have irregular shapes.

The community character framework is designed to satisfy the Shaw-Miller neutrality requirement. The character vector $\mathbf{c}(t)$ contains no racial composition data. The signals in M.1–M.7—occupational mix, land use category, housing tenure, commuting orientation, topographic zone, administrative zone co-membership, and transit accessibility—are all geographic and socioeconomic in nature. They may correlate with racial composition, as virtually any geographic characteristic does, but the correlation is incidental to the signal’s purpose, not definitional to it.

A plan drawn with community character weights can be defended on the following grounds: the algorithm is unaware of the racial composition of any tract; the weights encode measurable, publicly documented geographic and administrative features; and the formula is deterministic and verifiable by any party with access to the underlying data. No human judgment about the racial character of a neighborhood enters the computation.

3.3 State Constitutional Criteria

Four states with independent redistricting commissions provide the clearest statutory expressions of the communities of interest doctrine.

California. Proposition 11 (2008), which established the Citizens Redistricting Commission, requires that after compliance with federal law, the Commission shall adopt a plan that “shall favor keeping together communities of interest” (Cal. Gov. Code §8252(c), as enacted by Proposition 11 (2008)). A communities of interest is defined as “a contiguous population which shares common social and economic interests that would be best served by common representation.” The definition expressly excludes party affiliation and voting history.

Colorado. Article V, Section 44 of the Colorado Constitution directs the Independent Legislative Redistricting Commission to preserve “communities of interest, including but not limited to ethnic, cultural, economic, trade area, geographic, and demographic factors” as a mandatory criterion, ranked after equal population, VRA compliance, and contiguity. The enumeration of “trade area” and “geographic factors” maps directly to the M.4 commuting shed signal and the M.5 topographic signal.

Arizona. The Arizona Independent Redistricting Commission, established by Proposition 106 (2000), must give “communities of interest” weight as one of six criteria (Ariz. Const. Art. IV, Pt. 2, §1(14)). The Arizona Commission has interpreted the criterion to include “geographic, social, cultural, ethnic, and economic interests,” consistent with the M-track taxonomy.

Washington. RCW 44.05.090 instructs the Washington State Redistricting Commission to consider “communities of interest including racial and language minority groups.” This formulation is notable for explicitly including racial and language minorities alongside other community types, suggesting that community membership—rather than racial classification as such—is the organizing concept.

3.4 The Property Tax Fiscal Bond

Among the legally recognized instantiations of community of interest, the property tax fiscal bond is the strongest and most judicially familiar. Two households in the same school district, county subdivision, or municipal utility district do not merely share a geographic location; they are legally bound together as co-owners of a fiscal commons. They vote on the same bond measures, pay taxes to fund the same schools and fire stations, and receive services from the same institutions.

The Supreme Court recognized the civic significance of school district boundaries in *Milliken v. Bradley*, 418 U.S. 717 (1974) [U.S. Supreme Court, 1974], which declined to impose cross-district school desegregation remedies absent evidence that district lines were drawn for segregatory purposes. The Court’s analysis treated school district boundaries as presumptively legitimate community boundaries, reflecting genuine differences in local governance, tax policy, and civic identity. *Reynolds* itself acknowledged that “the integrity of various political subdivisions” is a legitimate state interest that may justify minor deviations from strict population equality.

The M.6 signal (administrative zone co-membership) operationalizes this doctrine by treating shared school district, fire district, water district, electric utility service territory, and county subdivision membership as direct measures of civic co-membership. Tracts that share all available zone types are, in the most literal administrative sense, members of the same community; the zone co-membership score encodes this fact as a continuous edge weight modifier.

3.5 The “Quantitative Neutral” Defense

An important practical virtue of the community character framework is that it provides a *quantitative neutral defense* against gerrymandering claims. When a litigant challenges a district boundary as arbitrary or politically motivated, the map-drawer using community character weights can respond:

- The boundary follows the edge of minimum community similarity, computed from publicly available data.
- The formula is $w(u, v) = w_{\text{base}} \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v)$, deterministic and fully disclosed.
- No map-drawer chose which communities to favor; the algorithm chose the geometrically cheapest cut, which is the cut between dissimilar communities.
- The formula can be replicated by any party with access to the same public data.

This is the redistricting analogue of the quantitative defenses that courts have found persuasive in employment discrimination cases: a facially neutral criterion, applied consistently, documented in advance, and verifiable by all parties, is presumptively lawful even if it produces disparate geographic effects.

4 Spectrum of Signals: M.1–M.7

Papers M.1 through M.7 each implement one community character signal. Table 1 provides an overview of all seven signals: their data source, the similarity metric used, geographic availability, and the legal precedent most directly supporting their use as a community of interest proxy.

Table 1: Spectrum of community character signals across the M-track. All sources are free and publicly available from federal agencies. Coverage refers to fraction of US states with usable data.

Paper	Signal	Data Source	Similarity	Coverage
M.1	Economic character	LODES WAC	Cosine	100%
M.2	Land use mix	NLCD	Cosine	100%
M.3	Housing character	ACS	Cosine	100%
M.4	Commuting shed	LODES OD	Jaccard	100%
M.5	Topographic zone	USGS 3DEP	Valley/ridge	100%
M.6	Zone co-membership	TIGER/EIA 861	Count ratio	100% ^a
M.7	Transit accessibility	GTFS	Isochrone overlap	~70%

^a School district + county subdivision + electric utility provide 100% national coverage. Fire and water districts cover approximately 60% and 40% of states, respectively.

4.1 M.1 — Economic Character (LODES WAC)

The Longitudinal Employer-Household Dynamics (LEHD) Origin-Destination Employment Statistics (LODES) Workplace Area Characteristics (WAC) file provides, for each census block, counts of jobs in 20 industry categories, 3 earnings bands, and 8 worker age groups [U.S. Census Bureau, Center for Economic Studies, 2022]. Aggregated to the tract level, this produces an occupational character vector reflecting where a tract fits in the regional economy: a manufacturing district, a financial services node, a retail corridor, a hospital campus. Two tracts with similar industry mixes are, economically, members of the same community of interest. Cosine similarity of the industry-share vector is the natural measure: it rewards proportional similarity of economic composition regardless of the absolute scale of employment.

Economic communities of interest are explicitly recognized by Colorado’s constitution (“trade area” factors) and have been accepted as a valid community type in Arizona IRC proceedings. LODES data is annual, free, and available from the Census Bureau’s LEHD program.

4.2 M.2 — Land Use Mix (NLCD)

The National Land Cover Database (NLCD), produced by the Multi-Resolution Land Characteristics Consortium, classifies every 30-meter pixel in the United States into one of 20 land cover categories [Multi-Resolution Land Characteristics Consortium, 2019]. Aggregated to the tract level, the NLCD produces a land use composition vector (fraction of developed land, fraction of forest, fraction of agricultural land, fraction of wetland, etc.) that captures the physical character of the tract. Two urban tracts are similar; two agricultural tracts are similar; an urban tract and an agricultural tract are dissimilar. The NLCD is updated approximately every five years and is freely available.

Physical geography as a community character signal has ancient legal roots. Natural features—rivers, mountain ranges, valley floors—have served as district boundaries since the colonial era, and state constitutions in many western states explicitly mention “natural boundaries” as a criterion. The NLCD operationalizes physical character systematically, across the entire landscape, without requiring the map-drawer to identify specific named features.

4.3 M.3 — Housing Character (ACS)

The American Community Survey (ACS) five-year estimates provide, at the tract level, counts of housing units by structure type (single-family, multi-family, mobile home), tenure (owner-occupied, renter-occupied), vacancy status, and year built [U.S. Census Bureau, 2020]. The housing character vector captures the residential identity of a tract: an established owner-occupied suburb of single-family homes built in the 1960s is a different community from a high-density rental market with recently constructed apartment buildings. Cosine similarity of the housing character vector captures community similarity in the sense of “neighbors who share a stake in the same type of housing market.”

Housing character is particularly relevant to the communities of interest doctrine because housing policy—zoning, rent control, historic preservation, homeowners association governance—is a primary vehicle through which neighbors exercise collective civic interests. Homeowners in the same zoning district share an interest in land-use decisions; renters in the same rental market share an interest in rent stabilization policy.

4.4 M.4 — Commuting Shed (LODES OD)

The LODES Origin-Destination (OD) file records, for each pair of census blocks with commuting flows, the count of workers who live in one block and work in another [U.S. Census Bureau, Center for Economic Studies, 2022]. Aggregated to the tract level, this produces a destination distribution for each residential tract: where do its residents go to work? Two tracts that send large fractions of their workers to the same employment centers are parts of the same functional economic area—the commuting shed or labor market area. Jaccard similarity of the destination set (tracts that receive at least one commuter from each origin) captures this relationship.

The commuting shed is a well-established concept in regional science and is recognized as a proxy for economic community of interest by California’s commission guidance and Colorado’s “trade area” criterion. Functional labor markets are a natural unit of democratic representation: residents who commute together to the same city have shared interests in transportation infrastructure, highway policy, and regional economic development.

4.5 M.5 — Topographic Zone (USGS 3DEP)

The USGS 3D Elevation Program (3DEP) provides one-meter to one-third-arc-second digital elevation models covering the contiguous United States [U.S. Geological Survey, 2022]. From elevation data, the topographic zone of a tract can be classified as valley floor, upland plateau, ridge, or slope zone using standard terrain analysis methods (slope curvature and hydrological drainage basin algorithms). Two tracts in the same valley share a drainage basin, a flood risk profile, an agricultural character, and often a historical settlement pattern; two ridge tracts share a different set of interests.

Topographic community signals are particularly relevant in western states, where valley communities in the Sierra Nevada, Cascades, Rockies, and Wasatch Range are economically and culturally distinct from adjacent upland areas. Natural boundaries have been cited approvingly in redistricting cases as evidence that a district boundary is not arbitrary [National Conference of State Legislatures, 2021].

4.6 M.6 — Administrative Zone Co-membership (TIGER/EIA 861)

Zone co-membership is the most legally direct community character signal. When two tracts share a school district, fire district, water district, electric utility service territory, and county subdivision, they are members of the same administrative community in every governmen-

tally recognized sense. The zone score

$$\text{zone_score}(u, v) = \frac{|\{z \in Z : \text{zone}_z(u) = \text{zone}_z(v)\}|}{|Z|} \quad (5)$$

counts how many of the available zone types are shared, normalized to $[0, 1]$. The zone co-membership formula replaces cosine similarity in equation (4) and satisfies all required properties: symmetry, $[0, 1]$ range, and graceful degradation to w_{base} when no zone data is available. Data for school districts and county subdivisions comes from TIGER/Line shapefiles (100% national coverage); electric utility service territories come from EIA Form 861 (100% coverage); fire and water districts from TIGER/Line Special District files (~60% and ~40% coverage, respectively). M.6 provides full documentation of the zone co-membership algorithm and its legal grounding.

4.7 M.7 — Transit Accessibility (GTFS)

General Transit Feed Specification (GTFS) data, published by transit agencies and aggregated by Transitland and the National Transit Database [Federal Transit Administration, 2022], encodes transit routes, stops, and schedules for approximately 70% of the US by population. Transit accessibility can be measured as the set of destinations reachable within a fixed travel time (e.g., 45 minutes) from a tract centroid; two tracts with overlapping reachable-destination sets are members of the same transit-accessible community. Isochrone overlap—the fraction of each tract’s reachable set that is also reachable from the other tract—provides a natural similarity measure.

Transit co-membership is a recognized community of interest in California’s commission guidance and in urban planning literature on transit-oriented development. In dense urban areas, transit accessibility is a primary determinant of where residents can work, shop, and access services, making it a more meaningful community bond than road network proximity alone. M.7 is limited to the approximately 70% of states and counties with published GTFS feeds; the algorithm degrades gracefully to the zone co-membership or geographic baseline for jurisdictions without transit data.

4.8 M.8 — Composite Community Index (preview)

Paper M.8, not yet published, will define a composite community character index that combines M.1–M.7 signals with expert-tunable weights. The composite is a weighted average of the individual similarity scores, providing a single number that can be used as the edge weight modifier when all signals are active. M.8 will also provide an expert witness template:

a standardized declaration format that maps the composite score to specific communities of interest recognized in prior redistricting proceedings.

5 Relationship to Existing bisect Weight Modes

5.1 The Three-Layer Compositor

The BISECT redistricting system structures every plan by three independently configurable components:

1. **Structure** (Layer 1): the bisection tree shape, which determines how the state is recursively divided into the required number of districts. Options include `standard-bisect`, `prime-factor` (ApportionRegions), and `ratio-optimal` (GeoSection).
2. **Weights** (Layer 2): the edge weight mode, which determines how costly each potential cut is for the METIS partitioner. Current modes are `geographic` (boundary length), `county` (county co-membership bonus), `unweighted`, and `vra-aligned`.
3. **Search** (Layer 3): the seed selection strategy, which determines which METIS result to accept when multiple random seeds are attempted. Options include `single`, `multi`, `convergence`, and `percentile`.

Community character weights occupy Layer 2. They do not alter the bisection tree structure (Layer 1) or the seed selection strategy (Layer 3).

5.2 Existing Weight Modes

The current weight modes are:

- **Geographic:** $w(u, v)$ proportional to the shared boundary length between tracts u and v . Long shared boundaries are more costly to cut; this encourages compact, round districts.
- **County:** a bonus weight applied to edges between tracts in the same county, as introduced in paper B.10. This encourages county-level community preservation without explicitly encoding any economic or cultural signal.
- **Unweighted:** $w(u, v) = 1$ for all edges. Used as a baseline; METIS minimizes edge count (number of tract boundary pairs split) rather than weighted cut cost.

- **VRA-aligned:** a weight mode that boosts edges between tracts with high minority population concentrations to preserve VRA Section 2 majority-minority configurations. See paper B.14 for details.

5.3 Alpha-Blending Community Character with Base Weights

The M-track introduces community character weights as an additional modifier that composes with any base weight mode via *alpha-blending*:

$$w_{\text{final}}(u, v) = \alpha \cdot w_{\text{base}}(u, v) + (1 - \alpha) \cdot w_{\text{char}}(u, v), \quad \alpha \in [0, 1], \quad (6)$$

where $w_{\text{base}}(u, v)$ is the weight from the active base mode (geographic, county, or unweighted) and $w_{\text{char}}(u, v) = w_{\text{base}}(u, v) \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v)$ is the community character modified weight.

When $\alpha = 1$, the final weight equals the base weight (community character is inactive). When $\alpha = 0$, the final weight is entirely determined by community character similarity. The default in M-track configurations is $\alpha = 0.5$, which gives equal weight to geographic boundaries and community character. The alpha parameter is tunable per plan configuration in the YAML config:

algorithm:

```
weights: geographic
character_signal: zone-membership
character_alpha: 0.5
```

5.4 CLI Integration

Each M-track signal is activated by the `--weights-override` flag with the corresponding signal name:

```
bisect state --state NC --year 2020 \
  --weights-override zone-membership \
  --version v_zone_m6
```

Multiple signals can be composed using the composite mode (M.8):

```
bisect state --state NC --year 2020 \
  --weights-override composite-community \
  --character-signals economic,zone-membership,housing \
  --version v_composite_m8
```

5.5 Interaction with Structure and Search Layers

Community character weights interact with the structure layer in the following way: the weights are computed once for the full state adjacency graph and remain constant throughout all bisection rounds. When the bisection tree recursively splits a subregion, METIS receives the subgraph of that subregion together with its original edge weights; no recomputation of character similarity is performed. This means that the character weights influence not just the first bisection but every subsequent level of the recursive partition.

Community character weights do not interact with the search layer (Layer 3). All seed selection strategies—single, multi, convergence, percentile—operate on the same weighted graph. The choice of search strategy affects how many METIS trials are run and which result is selected, but it does not alter the weights presented to METIS.

5.6 Composability with County Weights

The county weight mode (B.10) and the M.6 zone co-membership mode are related but distinct. County co-membership is a binary signal: two tracts are either in the same county or not. Zone co-membership (M.6) is a graded signal: tracts can share zero, one, two, three, or all available zone types. County subdivision membership (a dimension of M.6) is strictly more refined than county membership in New England states, where the town is the primary administrative unit and the county is a secondary one.

The two modes can be run independently or composed. Running zone co-membership as the base mode subsumes the county signal when county subdivision data is available, because county subdivision co-membership implies county co-membership in every state. In non-New England states where county is the primary unit, the county mode and the M.6 county-subdivision dimension produce nearly identical results.

6 Conclusion

The redistricting criterion “preserve communities of interest” is, at present, a phrase that courts accept and litigants invoke but that no judge can quantitatively enforce and no algorithm can natively satisfy. This paper has proposed a mathematical framework that changes that. By representing community character as a vector of tract-level signals, by measuring community similarity as cosine similarity of those vectors, and by encoding similarity as an METIS edge weight modifier, the framework translates a qualitative legal criterion into a deterministic, verifiable, partisan-neutral computation.

The framework satisfies the key legal requirements for defensible redistricting criteria. It is neutral under *Shaw v. Reno* and *Miller v. Johnson*: no racial, partisan, or incumbency data enters the character vector. It is consistent with the communities of interest doctrine as articulated in *Reynolds v. Sims* and as codified in California, Colorado, Arizona, and Washington state law. It is quantitative: the formula $w(u, v) = w_{\text{base}} \times \text{sim}(\mathbf{c}_u, \mathbf{c}_v)$ is fully disclosed, replicable by any party with access to the same public data, and immune to the charge that community identifications were made ad hoc for political advantage.

6.1 Expert Witness Role

The framework is designed for use by expert witnesses in redistricting litigation. A cartographer or political scientist serving as an expert witness can testify:

- The district boundaries were placed by an algorithm that minimizes cuts through edges connecting tracts of high community similarity.
- Similarity was computed from [NLCD / LODES / TIGER / EIA 861] data, which is publicly available and was processed by a deterministic formula.
- The formula does not incorporate, and is not aware of, the racial composition, partisan registration, or prior voting history of any tract.
- The communities preserved by the algorithm correspond to communities recognized in prior redistricting litigation and commission proceedings as legitimate communities of interest: school district co-members, commuting-shed partners, economic sector colleagues.

This template, developed in the context of M-track analysis, provides a standard framework for the expert declaration that would accompany any plan submitted to a court or commission.

6.2 Relationship to M.1–M.8

M.0 is the theoretical reference; M.1–M.7 are the empirical papers. Each of M.1–M.7 inherits the framework defined here—character vectors, cosine similarity, edge weight modifier, alpha-blending—and specializes it to one data source. Each paper validates the specialization by running BISECT on a selection of states and measuring the effect of the character weights on plan outcomes: compactness, school district integrity, administrative zone preservation, and proportionality gap. M.8 will compose all seven signals into a single composite index and provide an end-to-end expert witness template.

6.3 Open Questions

Three open questions are not addressed in this paper and are left for future work:

1. **Optimal α :** The alpha-blending parameter α is currently a fixed configuration choice. A data-driven method for selecting α to maximize community preservation subject to compactness constraints would be valuable.
2. **Signal orthogonality:** The M.1–M.7 signals are not independent; for example, economic character and commuting shed are correlated. A principal component analysis of the joint character matrix could identify a smaller set of orthogonal community dimensions.
3. **Temporal stability:** Community character changes over time. Whether a weight regime calibrated on the 2010 census remains appropriate for the 2020 redistricting cycle is an empirical question not addressed here.

The community character framework proposed in this paper provides a rigorous, legally defensible, and computationally tractable answer to the question that has long vexed redistricting reform: what, precisely, is a community of interest, and how do you know whether you have preserved one?

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